### Filter Data

Select Exam Score Range

55 101

Attendance_Percentage (select multiple)

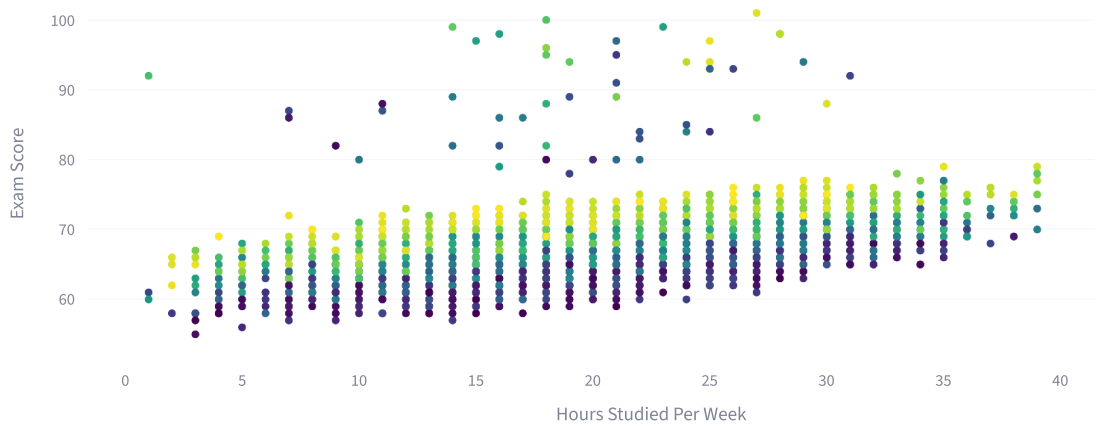
Choose options

# Student Performance Data Explorer

This app provides an interactive exploration of student performance data, and what factors influence student exam scores.

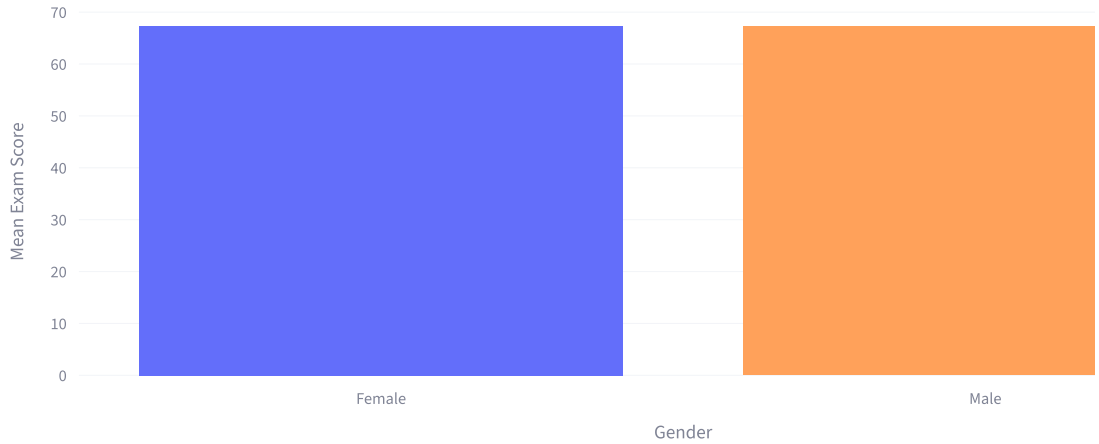
## Scatter Plot: Exam Score vs. Hours Studied Per Week (%)

**Exam Score vs. Attendance (%)**




## Bar Chart: Mean Exam Score by Gender

**Mean Exam Score by Gender**




The scatter plot above shows the relationship between hours studied per week and exam scores, with different colors representing different attendance percentages. The bar chart displays the mean exam scores for each gender, providing insights into performance differences based on gender. These visualizations help in understanding how attendance and amount of study hours impact exam scores, as well as gender-based performance differences. It can be observed that students with higher attendance percentages and higher hours studied per week tend to have higher exam scores. The average exam scores between genders are also similar, indicating that gender may not be a significant factor in exam performance. This dashboard was built using Streamlit and Plotly for interactive data visualization. The two interactive

elements, the score range slider and attendance percentage multiselect, allow users to filter the data on the scatter plot and explore different subsets interactively. The data shown when hovering over each gender provides a concise summary of the mean exam scores for that gender.